Nested Association Mapping (NAM) Reveals Genetic Architecture of Agronomic Traits in Quinoa

Reviewer 1

The manuscript presents a comprehensive and well-designed study aimed at dissecting the genetic architecture of key agronomic traits in quinoa through the integration of nested association mapping (NAM), genome-wide association studies (GWAS), and genomic prediction. The topic is timely and highly relevant, considering the increasing global interest in quinoa as a climate-resilient crop with exceptional nutritional value.

Response: Dear Reviewer, thank you for your comments and for your positive evaluation of our work. We appreciate the time and effort you devoted to reviewing our manuscript. Please find our detailed response to your question below.

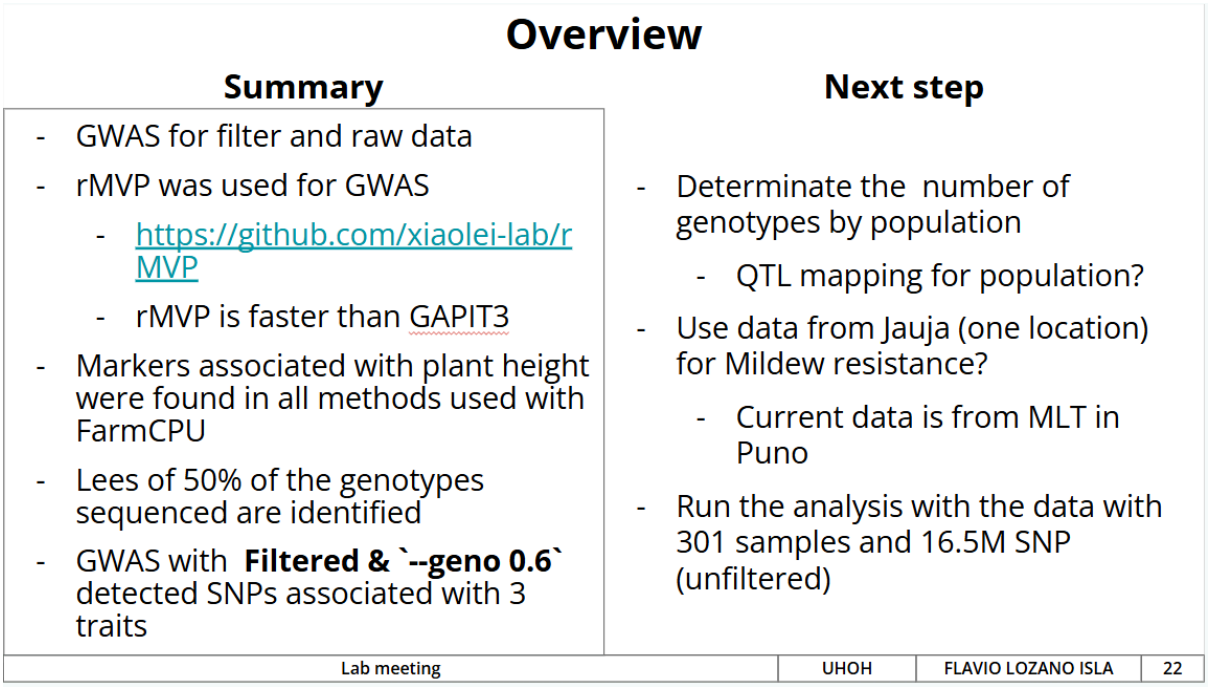
Comment 1: I have only one minor question for the authors. Did the authors compare the performance of different GWAS models implemented in GAPIT (e.g., GLM, MLM, CMLM, MLMM, BLINK, or others) before selecting FarmCPU? If so, it would be valuable to briefly explain the criteria used for choosing FarmCPU as the primary model and whether the significant loci identified by FarmCPU were consistent with those detected by other models.

Response 1: Thank you for this suggestion. We indeed performed several comparisons before selecting the final GWAS framework.

First, we compared the performance of two GWAS software packages, **GAPIT** (Lipka et al., 2012; Wang & Zhang, 2021) and **rMVP** (Yin et al., 2021). Both implementations produced virtually identical association results, identifying the same significant loci across the evaluated traits. Because rMVP provides faster computation and supports parallel processing, it was selected for the final analysis. A comparison of the outputs obtained from GAPIT and rMVP was previously presented during our laboratory meeting (attached figure).

We also evaluated the **NAM** R package, which is specifically designed for Nested Association Mapping populations. However, under our dataset and analytical conditions, this package identified fewer significant associations and showed lower consistency compared with the rMVP implementation. These comparisons are presented in the manuscript at **Supplementary Figures S6 and S7**.

Within rMVP, we further compared the performance of different GWAS models. Among them, **FarmCPU** consistently identified the same significant SNPs across multiple related traits while providing an effective balance between statistical power and control of false-positive associations. This performance is consistent with previous studies demonstrating that FarmCPU improves statistical power by iteratively fitting fixed and random effect models, thereby reducing confounding between testing markers and kinship while minimizing both false positives and false negatives (Liu et al., 2016). For these reasons, FarmCPU was selected as the primary GWAS model for the analyses presented in the manuscript.



References:

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Wang, J., & Zhang, Z. (2021). GAPIT Version 3: Boosting power and accuracy for genomic association and prediction. Genomics, Proteomics & Bioinformatics, 19(4), 629–640. https://doi.org/10.1016/j.gpb.2021.08.005

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